



ELECT MAINT Department
SIMHADRI

Note No :1

Ref:SMTTP/ELECT MAINT/2023-24/SWYDESP/941250

Date:

24/02/2024(18:23:38)

Subject:Review of LMI Prevention of Hydrogen Explosions due to hydrogen Leak in Generator Bus Ducts

" Prevention of Hydrogen Explosions due to hydrogen Leak in Generator Bus Ducts" is due for review in feb 2024. During the review following changes are envisaged serial number 5.0, 6.0,7.0 added inline with OGN, in addition to overhauling checklist, H2 leakage arresting checklist is added and made revision-2 from current LMI revision-1.

Approving authority for Reviewing of LMI lies with HOP.

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Note No:2

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Note No:3

LMI reviewed and found in line with OGN.

May be approved

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Note No:6

Pl review the LMI

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Note No:7

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Note No:8

May check the LMI and put up for approval

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Note No:9

Pls. check.

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Note No:10

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Note No:11

Pl discuss

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Note No:12

Discussed

LMI is checked in consultation with EMD, It May be approved.

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Note No:13

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Prevention of Hydrogen Explosions due to hydrogen Leak in Generator Bus Ducts.

**NTPC LIMITED
SIMHADRI THERMAL POWER STATION**

LOCAL MANAGEMENT INSTRUCTION
LMI /OGN/SYST/006
ISSUE NO.: 1, REVISION-2, DATE: Feb 2024

Prevention of Hydrogen Explosions due to Hydrogen Leak in Generator Bus Ducts.

APPROVED FOR

IMPLEMENTATION by _____
CHIEF GENERAL MANAGER (SIMHADRI)

DATE _____

LMI /OGN/SYST/006, ISSUE: 1, REVISION: 2, DT Feb 2024

Prevention of Hydrogen Explosions due to hydrogen Leak in Generator Bus Ducts.

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NTPC LTD. SIMHADRI THERMAL POWER STATION

OPERATION AND MAINTENANCE DEPARTMENT

01. TITLE OF LMI	Prevention of Hydrogen Explosions due to Hydrogen LeakGenerator Bus Ducts	
02. A) DOCUMENT CODE B) REVISION/ISSUE NO. & DATE	LMI /OGN/SYST/006 ISSUE NO.: 1, REVISION-2, DATE: Feb 2024	
03. ORIGINAL/REF. DOCUMENT	LMI /OGN/SYST/006 Rev 0,1999	
04. PREPARED BY	Sr. Manager (EM)	Anil Kumar Pinninti
05. CHECKED BY	DGM (EM)	Meda Nagaraja
06. REVIEWED BY	HOD(EM)	D.K Tomar
07. REVIEWED BY	GM(Operation)	T. Rajagopalan
08. REVIEWED BY	GM(O&M)	Suresh John David

Prevention of Hydrogen Explosions due to hydrogen Leak in Generator Bus Ducts.

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Prevention of Hydrogen Explosions due to hydrogen Leak in Generator Bus Ducts.

1.0 INTRODUCTION:

There are many routes of escape of hydrogen from the generator casing. One of them could be in the bus duct area. Because hydrogen becomes flammable and/or explosive at certain concentrations (4% to 74%), enough care must be taken to prevent hydrogen leaks. An inflammable mixture of hydrogen and air may rapidly develop, requiring only a static electrical discharge or contact to initiate an explosion. Incidences of hydrogen explosion have occurred and have caused huge damages to the operating units.

2.0 SCOPE:

The document covers the procedure to be followed for identifying the hydrogen leakages in generator bus duct unit – 1, 2, 3 & 4 and corrective actions to be taken thereof.

3.0 REFERENCE DOCUMENT:

Operation Guidance Note OGN/OPS/SYST/006.

4.0 SUPERSEDED DOCUMENTS: NIL

5.0 CAUSE OF OCCURRENCE

Hydrogen can leak through the bus-duct due to

- i) Failure of the bushing seals
- ii) Cracks getting developed in the bushings

Failure of the bushing seals can occur due to bad quality/improper fixing during replacement of bushings, and possibility of cracks getting developed can occur due to cumulative effects of system jerks. Once the gas leaks through the bus-duct area, the probability of explosion is quite high due to sudden fall in the hydrogen purity and getting the ignition source from overheated braids/flexibles. The probability of explosion inside the casing is quite less until and unless the hydrogen purity falls below the specified level. The bus duct explosions in generators can cause extensive damage. Basically, the leakage of hydrogen into the bus-duct can be quite dangerous because hydrogen purity deteriorates quite fast and ignition source through heated bus-duct flexibles/connections can trigger the explosion process.

6.0 PREVENTION OF OCCURRENCE DURING OPERATION

The occurrence can be prevented during operation of the unit by careful monitoring of the hydrogen cylinder consumption. Sudden increase of hydrogen cylinder consumption can be due to either internal or external hydrogen leakage. Internal leakages can be identified through the detection of gas bubbles in the gas trap whereas external leakages could be through the seals, gas

Prevention of Hydrogen Explosions due to hydrogen Leak in Generator Bus Ducts.

panels and gasket failures. Probability of explosion occurring is more through bus-duct where flexible links heating/sparking due to loosening or other reasons can provide necessary ignition source to the leaking hydrogen.

7.0 PREVENTION DURING MAINTENANCE OF UNITS

Braided flexible connections and bolted joints between the terminals of generators can deteriorate due to several factors. These are vibration, overheating of joints and damage caused during removal and replacement of various parts during capital overhauls. Problems have also been experienced with poor current distribution at braided connections, resulting in overheating of braids.

The temperature monitoring of the unit bus-duct through thermo-vision camera scanning periodically should be the essential feature that can foretell any abnormal heating taking place.

If the contact surfaces are not cleaned properly before assembly and bolts are not tightened to the correct torque setting, the problem of heating can occur. All braided flexible connections, laminated connections and bolted joints of generator bus-bar system should be inspected for signs of deterioration, overheating or major damage during normal maintenance outage.

8.0 SPECIAL INSTRUCTIONS TO BE FOLLOWED:

Unit wise daily consumption of Hydrogen is to be monitored and any excessive consumption will be intimated to electrical maintenance.

Instruction	Responsibility	Document Ref.
Hydrogen consumption Monitoring on daily basis.	Operation / Generator maintenance group	Shift In charge log book.
Identifying and Arresting of Hydrogen leakage	Generator maintenance group	LMI/ELEC-006

Prevention of Hydrogen Explosions due to hydrogen Leak in Generator Bus Ducts.

9.0 METHODOLOGY OF IMPLEMENTATION:

- H₂ cylinder consumption will be monitored carefully.
- If the consumption of cylinders per day is more than two, leakage is suspected. The source of leakage is to be identified with portable leakage detector.
- Observe for bubbles in the cooler vents / drains etc.
- If leakage is external, then carefully monitor the consumption of the cylinders and decision has to be taken for availing shutdown.
- After availing the shutdown, check for the leakages in the adopter panel.
- If leakage is from generator bushings / seals then replace them after purging out the H₂.
- After attending the H₂ leakage and after pressurization of H₂ check for any leakages in the adopter box before exciting of the machine.

10.0 RESPONSIBILITY CENTRE FOR IMPLEMENTATION:

Generator maintenance group for implementation and overall responsibility lies with EM-Lab (I/C)

11.0 RECORD KEEPING:

Records will be maintained by generator maintenance group in LMI/ELEC-006.

12.0 REVIEW:

HOP (Simhadri) would review the procedure every two years.

LMI REFERENCE : LMI /OGN/SYST/006

LMI DESCRIPTION: **Prevention of Hydrogen Explosions due to Hydrogen Leak in Generator Bus Ducts.**

**HYDROGEN LEAK IN GENERATOR BUS DUCT
OVERHAUL CHECK LIST**

UNIT:

DATE:

SL. NO	DESCRIPTION	REMARKS
1	Leakage in the Bushings	
2	Cracks in the Bushings	
3	Cleanliness of the bushings	
4	Bushing Flange gasket condition	
5	Damage/Distortion of Bushing flexible connectors	
6	Overheating of Bushing flexible connectors	
7	Discoloration of Bushing flexible connectors	
8	Looseness in bushing flexible connectors	
9	Tightness of bushing flexible connectors	
10	Cleanliness of mesh	

CHECKED BY:

LMI REFERENCE: LMI /OGN/SYST/006

**LMI DESCRIPTION: Prevention of Hydrogen Explosions due to Hydrogen
Leak in Generator Bus Ducts.**

Date: _____

1. UNIT:
2. H2 leakage Location:
3. SAP work order no:
4. Action Taken:
5. Condition after arresting leakage:

(Signature)

EM-LAB